

A retrospective evaluation of a COVID-era community engagement program

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Program: A community engagement alliance was formed across several US states during the COVID-19 pandemic

- **Cohort 1:** Funded November 2020
- **Cohort 2:** Funded April 2021
- By May 2021, vaccines widely available for 16+ in US

What we know:

- **Counties**
- **Vaccination rates**

How to evaluate the program?

- **Hypothesis:** Counties where a community engagement program was implemented had **higher COVID-19 vaccination rates**.

Based on:

- **Outcome of interest**
- **Causality after the fact**

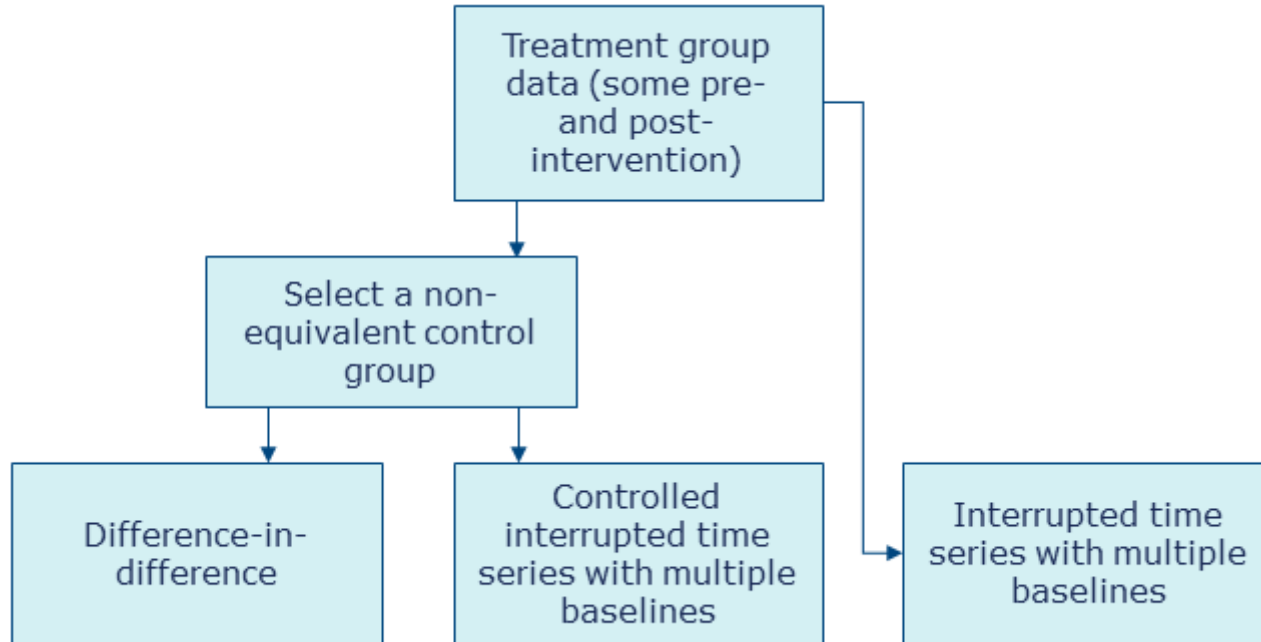
Quasi-Experimental Methods

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n}$$

$$\hat{V}(\bar{y}_w) = \frac{1}{K} \sum_{r=1}^R (\bar{y}_w^r)^2$$

$$R = 100, K = 100(1 - 0.3)^2$$

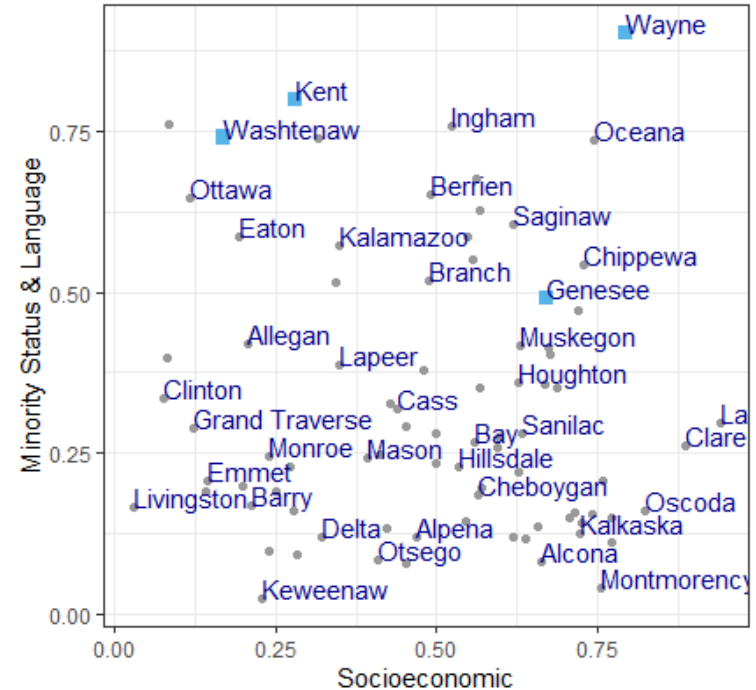
Quasi-experiments aim to evaluate interventions without randomization.



Non-equivalent control group

- Matched each treatment county to a control county by Euclidean distance
 - By **socioeconomic status, household composition, and minority status and language** from CDC's Social Vulnerability Index (SVI).
 - For states that targeted rural communities, we also used **population density**.
 - Candidates for the matched counties were in the **same state** and **not adjacent** to any intervention county.

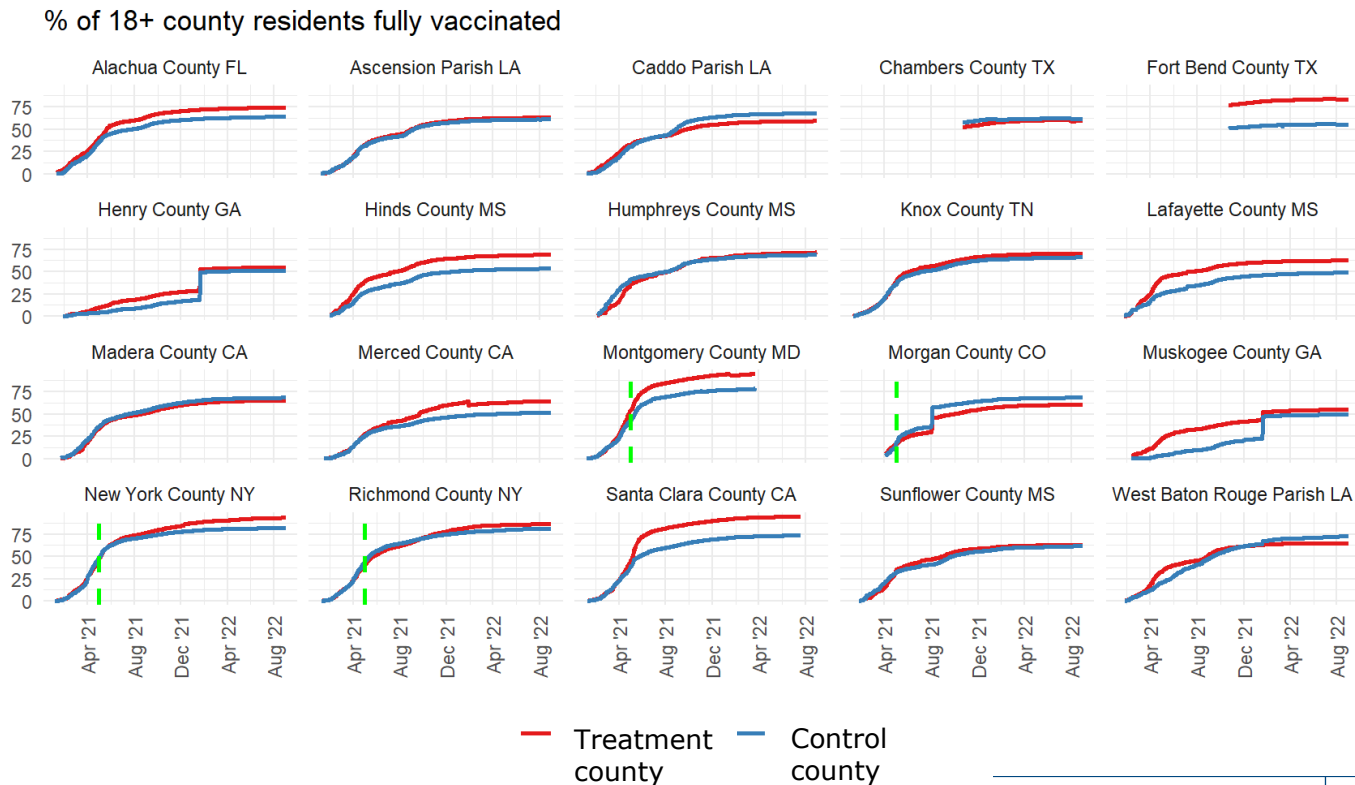
Michigan



Vaccination rates were obtained from the **CDC COVID Data Tracker** for matching pairs of counties

- 115 treatment counties

- 115 non-equivalent control counties

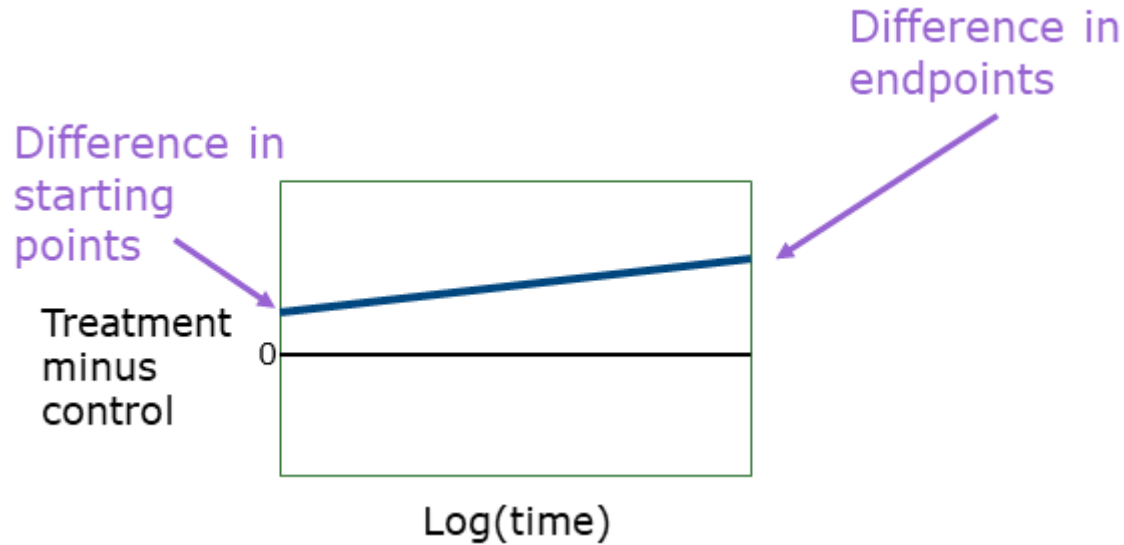


Models

	Model equation
Difference-in-difference (n=115 county pairs)	
115 counties, randomization	$(y_1^T - y_0^T) - (y_1^C - y_0^C)$
Controlled Interrupted Time Series with Multiple Baselines (n=627 time-county pairs)	
Quarterly, mixed model with covariates	$\begin{aligned} &y_t^T - y_t^C \\ &= \beta_0 + \beta_1 I(\text{funded}_t) + \beta_2 (X^T - X^C) \\ &+ \beta_3 \log(t) + \gamma_{\text{county}}^1 + \gamma_{\text{county}}^2 \log(t) + \epsilon_t \end{aligned}$
Interrupted Time Series with Multiple Baselines (n=627 time-county pairs)	
Quarterly, mixed model with covariates	$\begin{aligned} &y_t^T \\ &= \beta_0 + \beta_1 I(\text{funded}_t) + \beta_2 X^T + \beta_3 \log(t) \\ &+ \gamma_{\text{state}}^1 + \gamma_{\text{state}}^2 \log(t) + \epsilon_t \end{aligned}$

Difference-in-difference

Counterfactual is based on comparison between **treatment** and **control group** using one time point at the beginning and end of **post-intervention**

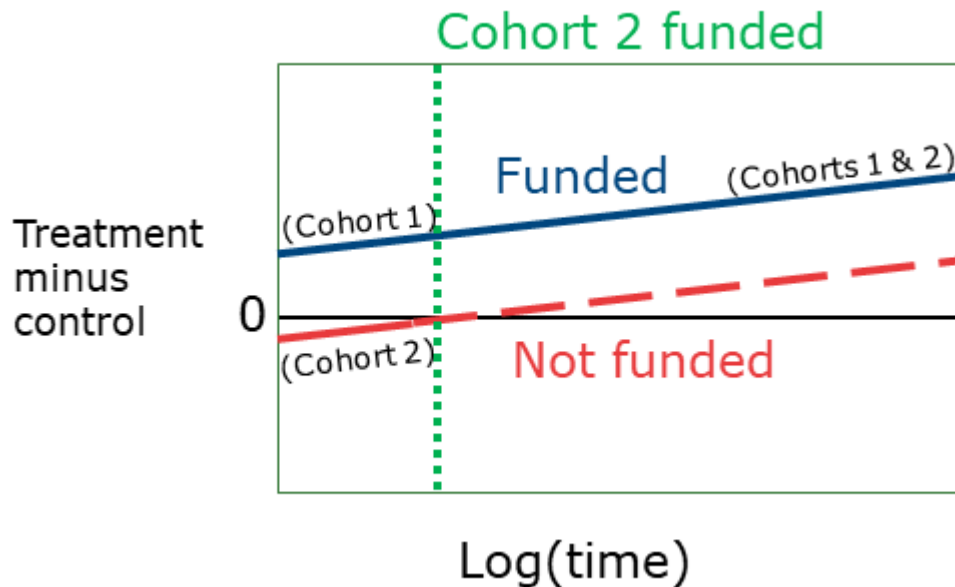


$$(y_1^T - y_0^T) - (y_1^C - y_0^C)$$

Controlled Interrupted Time Series Analysis with Multiple Baselines

Counterfactual is based on comparison to **control group** and **pre-intervention group**

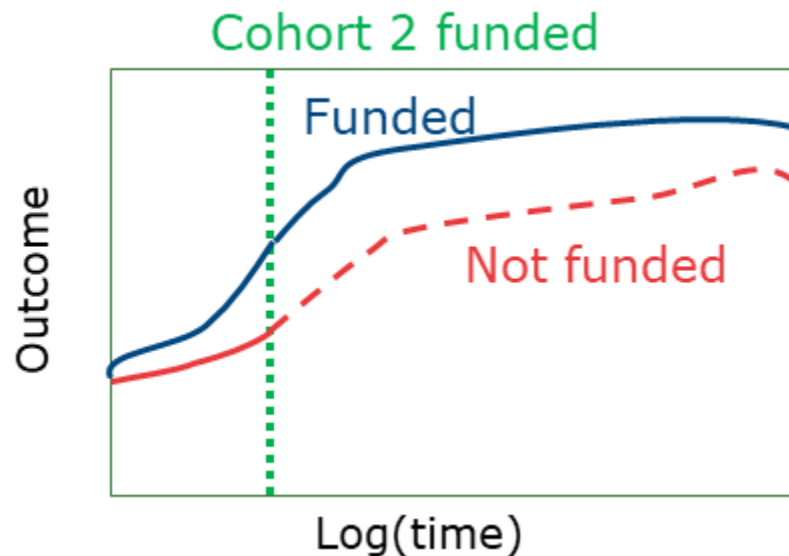
$$\begin{aligned} y_t^T - y_t^C &= \beta_0 + \beta_1 I(\text{funded}_t) \\ &+ \beta_2 (X^T - X^C) + \beta_3 \log(t) \\ &+ \gamma_{county}^1 + \gamma_{county}^2 \log(t) + \epsilon_t \end{aligned}$$



Interrupted Time Series Analysis with Multiple Baselines

Counterfactual is based on comparison to **pre-intervention group**

$$\begin{aligned} y_t^T &= \beta_0 + \beta_1 I(\text{funded}_t) + \beta_2 X^T \\ &+ \beta_3 \log(t) + \gamma_{state}^1 \\ &+ \gamma_{state}^2 \log(t) + \epsilon_t \end{aligned}$$



Results

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n}$$

$$\hat{V}(\bar{y}_w) = \frac{1}{K} \sum_{r=1}^R (\bar{y}_w^r)^2$$

$$R = 100, K = 100(1 - 0.3)^2$$

Assessing match balance



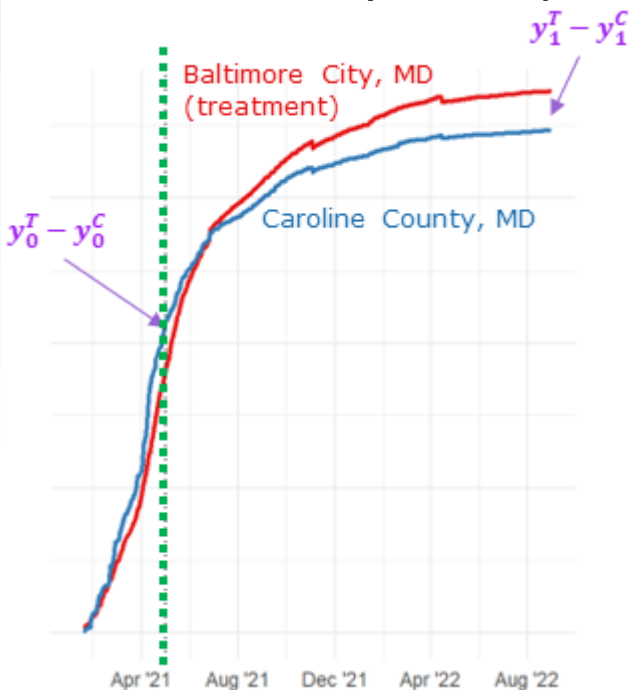
<0.1 indicates no major imbalance, but higher values are considered acceptable if **covariates** are considered in analysis

● SMD(Treatment, Match)
● SMD(Treatment, Nonmatch)

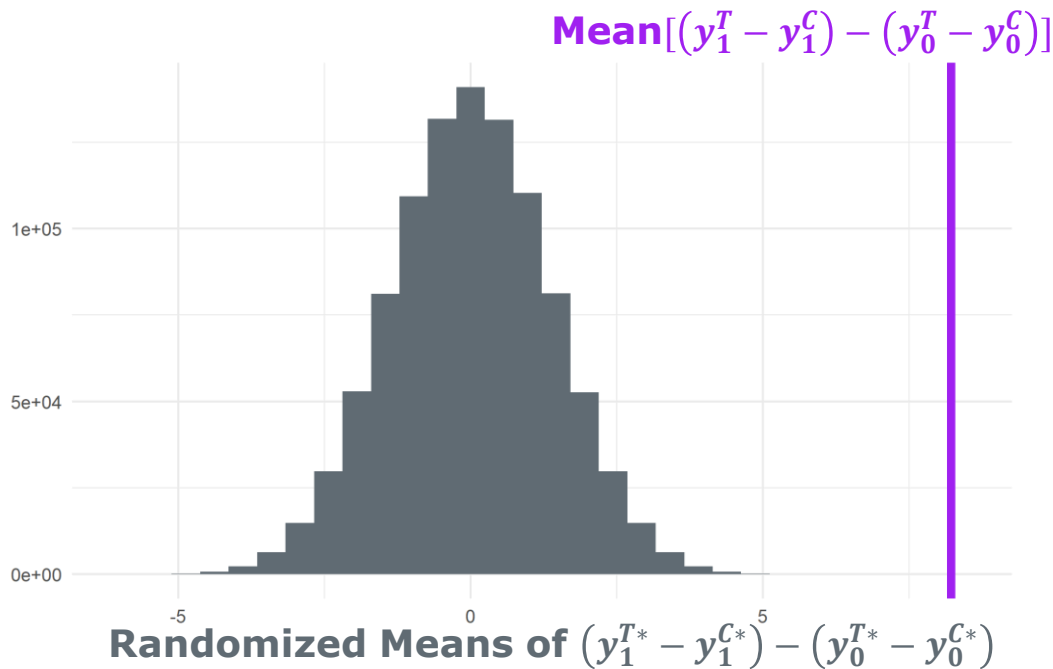
Both interrupted time series analyses included **covariates**
X

Difference-in-difference: vaccination rates were **8.2 points higher** in intervention counties

Matched county example:



Null distribution and test statistic:



Summary of Results

Method	Difference estimate	P-value
Difference-in-difference		
All counties	+8.2	<0.0001
Controlled Interrupted Time Series with Multiple Baselines		
Quarterly, mixed model with covariates	+5.9	<0.0001
Interrupted Time Series with Multiple Baselines		
Quarterly, mixed model with covariates	+27.4	<0.0001

Conclusions

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n}$$

$$\hat{V}(\bar{y}_w) = \frac{1}{K} \sum_{r=1}^R (\bar{y}_w^r)^2$$

$$R = 100, K = 100(1 - 0.3)^2$$

In Conclusion,

- All three models support the hypothesis that counties participating in the program had **higher vaccination rates**.
- The **definition of the counterfactual** impacted the results. For example, it is possible that the control group was also impacted by the intervention.
- More **comparisons** between quasi-experimental models are needed to understand the threats to internal validity in the context of public health intervention evaluation.

Thank You

For more information, please contact:

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Also see CEAL Program Evaluation Report, 2023



Presentation Citations

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$$\hat{V}(\bar{y}_w) = \frac{1}{K} \sum_{r=1}^R (\bar{y}_w^r)^2$$

$$R = 100, K = 100(1 - 0.3)^2$$

Citations

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<https://covid.cdc.gov/covid-data-tracker>. Accessed September 14, 2022.
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- **Jandoc**, Racquel, et al. "Interrupted time series analysis in drug utilization research is increasing: systematic review and recommendations." Journal of clinical epidemiology 68.8 (2015): 950-956.
- **Bernal**, James Lopez, Steven Cummins, and Antonio Gasparrini. "The use of controls in interrupted time series studies of public health interventions." International journal of epidemiology 47.6 (2018): 2082-2093.

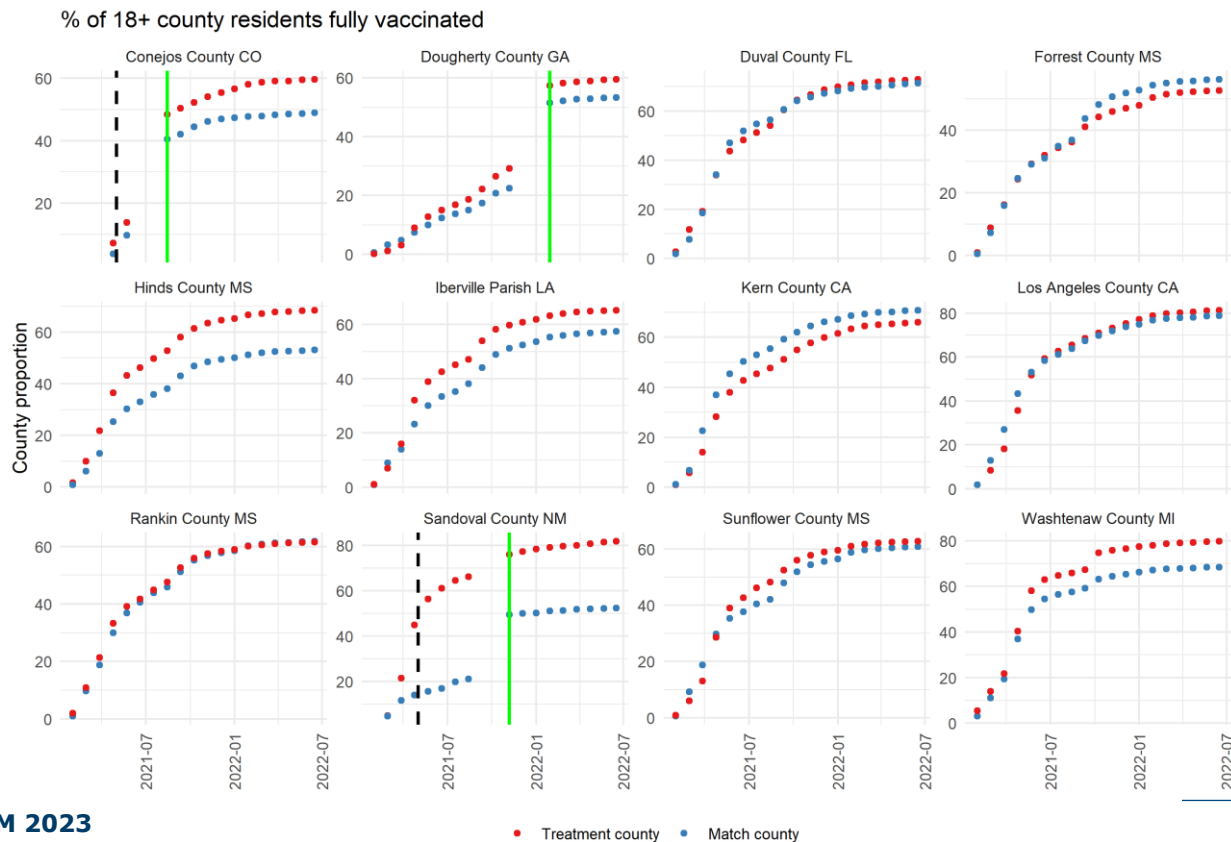
Appendix

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n}$$

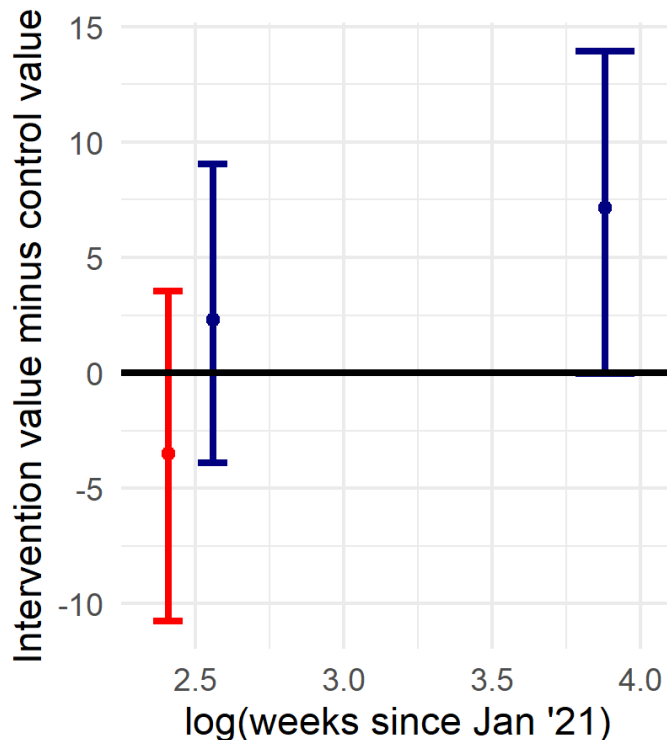
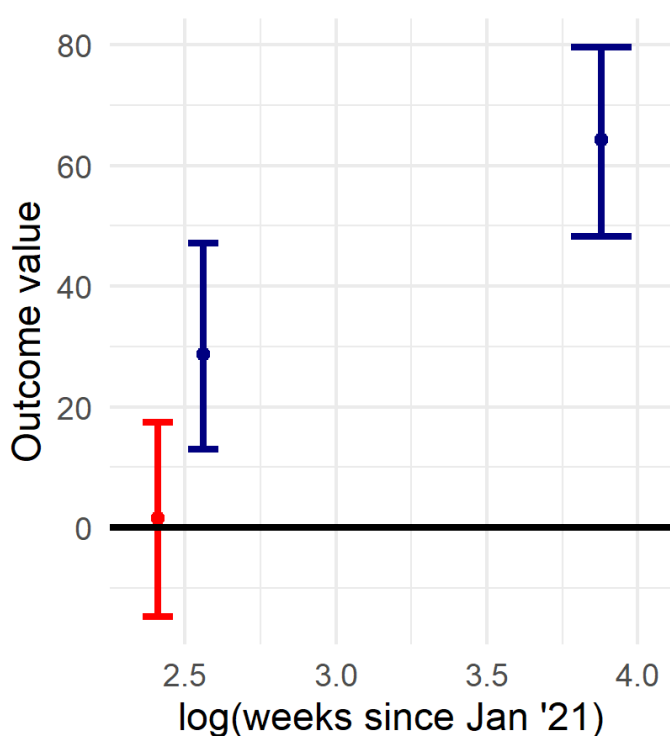
$$\hat{V}(\bar{y}_w) = \frac{1}{K} \sum_{r=1}^R (\bar{y}_w^r)^2$$

$$R = 100, K = 100(1 - 0.3)^2$$

Removed irregular data.



Interrupted Time Series Analysis with Multiple Baselines compared to **Controlled** version



Results

Method	Difference estimate	P-value
Difference-in-difference		
All counties	+8.2	<0.0001
Close matches	+6.6	<0.0001
Cohort 1	+6.1	<0.0001
Cohort 2	+14.0	<0.0001
Controlled Interrupted Time Series with Multiple Baselines		
Quarterly, mixed model	+5.7	<0.0001
Quarterly, mixed model with covariates	+5.9	<0.0001
Interrupted Time Series with Multiple Baselines		
Quarterly, mixed model with covariates	+27.4	<0.0001